MIRAI BANK PROJECT

Milestone 1 [B1]

User Interface and Operations Research Report

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The minimal operations the prototype must support have been organized into 6 core functions, based on the existing UI screen design (11 screens in total). In addition to the 5 example functions given in the acceptance criteria (connect, upload, view data information, register data, and manage access), "NFT generation," which is explicitly named in the Milestone Output description, has been added as an independent operation. 4

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1. Research Overview

This report is Milestone 1 ("Prototype research and basic specifications") Output B1, the User Interface and Operations Research Report. This prototype considers a minimal user interface for performing operations such as connecting to a wallet, uploading data, registering information, and generating NFTs.

This research builds on the screen structure and permission model already finalized in the Project Handover v3.0, and organizes the following two points. Detailed technical implementation (specific components, API details, etc.) will be addressed in the next milestone, M2 (Prototype Detailed Design); this document focuses as much as possible on plain-language explanations.

- **B1-1:**

  Organize the key user operations in a table (including at least 5 core functions)

- **B1-2**

  Illustrate the flow of data for each operation as a data flow diagram

2. List of Key User Operations (Corresponding to: B1-1)

The minimal operations the prototype must support have been organized into 6 core functions, based on the existing UI screen design (11 screens in total). In addition to the 5 example functions given in the acceptance criteria (connect, upload, view data information, register data, and manage access), "NFT generation," which is explicitly named in the Milestone Output description, has been added as an independent operation.

No	Operation	Description	Actor (Role)
1	**Connect (Wallet Connection)**	Connect to the user's wallet and use ownership of that wallet in place of identity verification. On first connection, a personal storage space is prepared automatically.	All users (not yet connected)
2	**View Data Information (List / Detail View)**	Display a list and details (status, registration progress, etc.) of the groups the user belongs to and the capsules created so far.	All users
3	**Register Data (Capsule Creation / Piece Registration)**	Create a new capsule and register content (a Piece), such as text or an image. For a group capsule, a registration slot is automatically prepared for each member, and the manager also sets the opening mode (who can open the capsule, and how).	All users (the creator becomes the capsule's manager)
4	**Upload (Encrypt / Store Content)**	Text or images to be registered are encrypted on the user's device before being sent, and the encrypted content is uploaded to storage (decentralized storage).	All users (the piece registrant)
5	**NFT Generation (Sealing)**	Finalize the registered content and issue a digital certificate (an NFT) proving that the content will not change, sealing the capsule.	The capsule's manager
6	**Manage Access (Permissions / Member Management)**	Add or remove group members, change roles, and configure who may view the content after it is opened. For a group capsule, this also covers a manager's mass opening or emergency opening, depending on the opening mode that was set.	The group's manager

2.1 Screen ID Mapping Table

This section organizes the correspondence between all 11 screens of the UI prototype and the 6 operations above.

Screen ID	Screen Name	Corresponding Operation	Main Role
SCREEN-01	QR Login	① Connect	All users
SCREEN-02	First-Time Registration	① Connect	All users
SCREEN-03	Dashboard	② View Data Information	All users
SCREEN-04	Capsule Creation	③ Register Data	All users
SCREEN-05	Manager Operations	② View Data Information	Group / capsule managers
SCREEN-06	Piece Registration	③ Register Data / ④ Upload	All users
SCREEN-07	Piece Approval Management	Addressed in M2 (Prototype Detailed Design)	The capsule's manager
SCREEN-08	NFT Confirmation	⑤ NFT Generation	The capsule's manager
SCREEN-09	Opening Viewer	Addressed in M2 (Prototype Detailed Design)	All users
SCREEN-10	Wallet Information	② View Data Information	All users
SCREEN-11	Group Management	⑥ Manage Access	The group's manager

3. Processing Flow for Each Operation (Corresponding to: B1-2)

Each key operation is organized from three perspectives: "what the operation is for," "what flow it follows," and "points to keep in mind." The diagrams are general-purpose flow diagrams showing the broad flow from the user's action to the confirmed result; detailed technical implementation is addressed in Milestone 2 (Prototype Detailed Design).

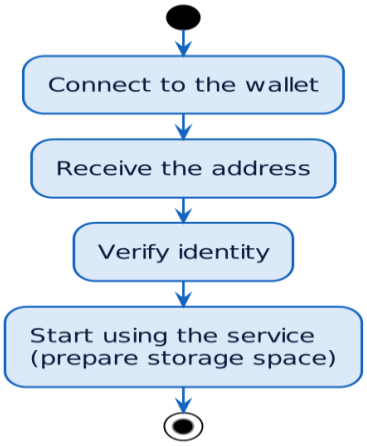
3.1 Connect (Wallet Connection)

Purpose:

This operation is the entry point through which all use of ChainCapsule begins. There is no need to create a new password; by design, the user's existing Cardano wallet is used directly as identity verification, allowing the user to start using the service immediately without any registration hassle.

Processing Flow:

When the user opens a wallet app (a CIP-30-compliant wallet such as Lace or Eternl) and approves the connection to Chain Capsule, the wallet returns an address and a signature. This signature confirms that "the owner of this address has, without doubt, approved the connection." The server verifies identity based on this address; if this is the user's first connection, a personal storage space is automatically prepared on the spot. On subsequent connections, the existing storage space is simply retrieved.



Point to Note: As a consideration for users who do not yet have a wallet, guidance on how to set up a wallet should also be considered for the first-connection flow. In addition, in environments with multiple wallet extensions installed, the user should be explicitly asked which wallet to use at connection time, so they are not left unsure which to choose.

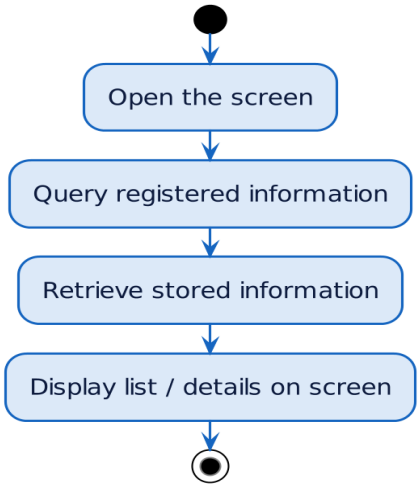
3.2 View Data Information (List / Detail View)

Purpose:

This operation lets users check the status of their own records at any time, with peace of mind. Being able to see a capsule's status (in progress, sealed, or opened) at a glance builds the user's sense of security and trust.

Processing Flow:

When the dashboard screen is opened, the app queries the server for the list of groups and capsules associated with that user. The server retrieves the relevant information from what is stored and returns it as a list, along with each capsule's status and registration progress. When the user selects a specific capsule, more detailed information is shown, such as the number of pieces registered and the planned opening date.



Point to Note: Because the content of a capsule (the text or photos themselves) can no longer be viewed once it is sealed, this screen should be designed to show only peripheral information such as status and progress, never the content itself, until the moment of opening.

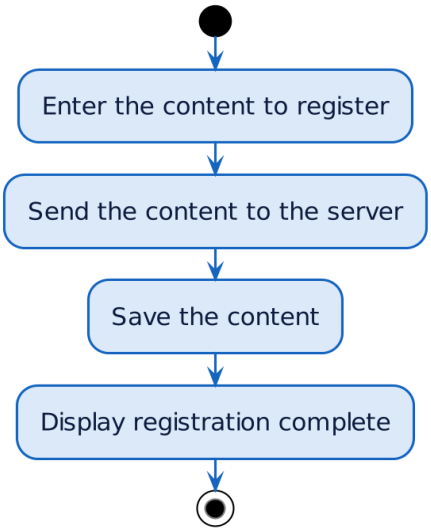
3.3 Register Data (Capsule Creation / Piece Registration)

Purpose:

This is ChainCapsule's central operation, the step where the user turns "something to leave for the future" into an actual, concrete form. The content registered here is what will later be sealed and protected until the specified date.

Processing Flow:

The user first creates a new capsule, entering basic information such as the planned opening date. For group use, the manager also selects the "opening mode" (who can open the capsule, and how) at this point. They then register the content to leave inside the capsule (a Piece) one at a time. A registration slot is prepared in advance for each participating member, and each member registers their own content into their own slot. The entered content is sent to the server and saved as a draft.Processing Flow:



Point to Note: While still a draft, the content should be editable and deletable any number of times, giving users the reassurance that they can freely rewrite it until they finalize (seal) it. For group use, it is also important for the manager to be able to see who has registered and who has not yet done so. Because the opening mode cannot be changed once the capsule is sealed, the screen should clearly explain what the choice means at the time it is made (see C1 for details on opening modes).

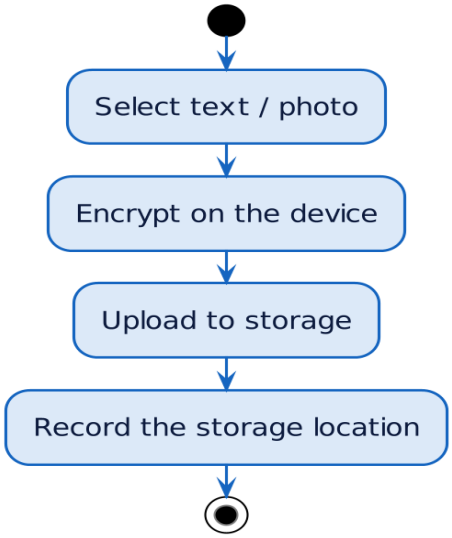
3.4 Upload (Encrypt / Store Content)

Purpose:

This operation safely and reliably stores the text or photos the user has registered. It is the technical foundation underpinning ChainCapsule's promise to "protect the content until the promised day."

Processing Flow:

When the user selects text or a photo, the content is encrypted on the user's own device before it is sent. The encrypted content is uploaded to storage (decentralized storage), and once it is successfully saved, information indicating where it was stored is recorded on the server. This storage-location information is later written into the certificate (the NFT) as well, when the capsule is sealed.



Point to Note: By encrypting on the device, the design ensures that no one — including the service operator — can see the content in transit or in storage. When handling large files such as photos or videos, a clear progress indicator for the upload should also be considered for the user.

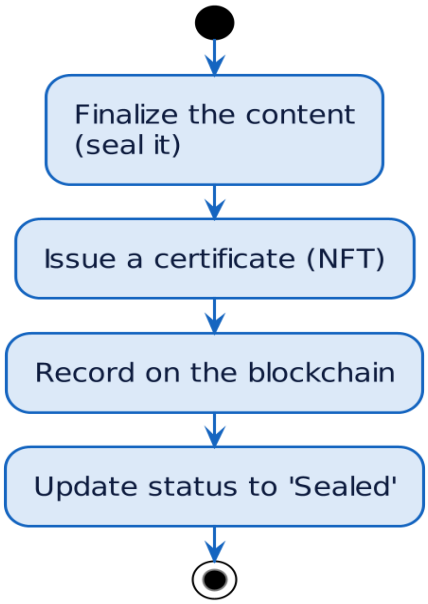
3.5 NFT Generation (Sealing)

Purpose:

This operation finalizes registered content into a state that "can no longer be changed," and proves that fact in a way that is unmistakably clear to everyone. It is the step that underpins ChainCapsule's trustworthiness at its core.

Processing Flow:

When the capsule's manager finalizes (seals) the content, preparations begin to issue a digital certificate (an NFT) proving that the content will not change. Once the manager signs with their wallet, the certificate is recorded on the blockchain and can no longer be rewritten by anyone. As a result, the capsule's status is updated to "Sealed," and no further additions or edits to the content are possible.



Point to Note: Because sealing cannot be undone, a confirmation screen must be provided beforehand, asking the user to make sure "this content is really final." For group use, it is also essential to confirm that all members have finished registering before allowing the capsule to be sealed.

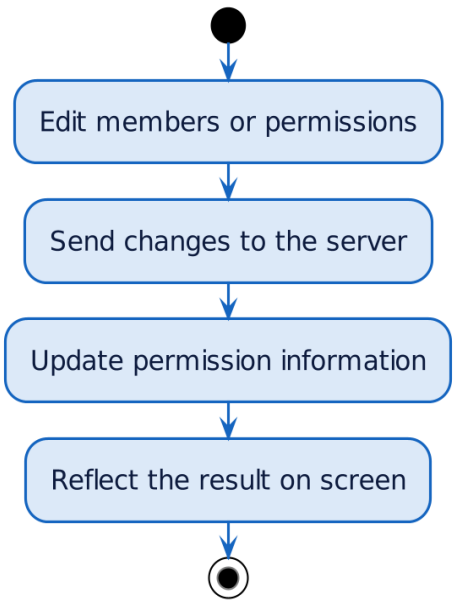
3.6 Manage Access (Permissions / Member Management)

Purpose:

This operation appropriately controls "who can do what" in group use, where multiple people share a single capsule. Alongside managing members and roles, it also covers who can view content, and how, according to the capsule's opening mode.

Processing Flow:

When the group's manager adds or removes members, or changes roles, on the group management screen, the changes are sent to the server and the permission information is updated. How opening works depends on the opening mode chosen at capsule creation. Under the basic mode, each member can only open and view the content they themselves registered. If mass opening by managers was chosen, managers can also open and view all of the content together once the planned opening date/time has arrived. If emergency opening was additionally chosen, managers can open the capsule even before the planned date/time, on the condition that a predetermined number of managers agree.



Point to Note: A mistake in permission management can lead to serious problems, such as content being visible to someone who should not see it, or being hidden from someone who should. Emergency opening in particular — where a manager can open the capsule before its planned date/time — is an exceptional capability that breaks the normal promise that "no one can open it before the specified date," so the design requires agreement from multiple managers, ensuring it can never be executed by a single manager's decision alone (see C1 for the detailed conditions).

4. Discussion on the Minimal UI Configuration

From the standpoint of the "minimal user interface" referred to in the Milestone Output description, this section organizes, among the 11 screens, which ones should have their basic specifications finalized within the scope of this document (M1), and which should be examined in more detail in the next milestone, M2 (Prototype Detailed Design).

Priority	Screen	Reason	Phase
Required	SCREEN-01 (QR Login) SCREEN-02 (First-Time Registration)	No operation can proceed without a wallet connection, so this is needed first	M1 (this document)
Required	SCREEN-04 (Capsule Creation) SCREEN-06 (Piece Registration)	These are the core "Register Data" and "Upload" operations	M1 (this document)
Required	SCREEN-08 (NFT Confirmation)	This is the core "NFT Generation" operation	M1 (this document)
Semi-Required	SCREEN-03 (Dashboard) SCREEN-05 (Manager Operations) SCREEN-10 (Wallet Information)	Corresponds to "View Data Information." Needed to verify behavior, but a simple version is sufficient	M1 (this document)
Phased	SCREEN-11 (Group Management)	Corresponds to "Manage Access." Priority is given to verifying individual use; only an overview of group functionality is covered here	Overview in M1 Detail in M2
Deferred to M2	SCREEN-07 (Piece Approval Management) SCREEN-09 (Opening Viewer)	More detailed design is needed for the approval procedure and the presentation at opening, so these are addressed in the next phase	M2

As shown above, the screens (SCREEN-01, 02, 04, 06, 08) corresponding to the four core operations (Connect, Register Data, Upload, and NFT Generation) are treated as top priority, with their basic specifications finalized in this document, while the group feature (SCREEN-11) is covered only at an overview level. Piece Approval Management (SCREEN-07) and the Opening Viewer (SCREEN-09) require more detailed design and are therefore addressed in Milestone 2. This approach aims to allow the full minimal-UI sequence (wallet connection → capsule creation → piece registration → NFT issuance) to be verified at an early stage.

5. Summary

In conclusion, this report defines the minimal user interface and operation model required for the first prototype phase of ChainCapsule. The key point of the proposed UI is that the user can complete the full basic sequence—connecting a wallet, creating a capsule, registering content, uploading encrypted data, and sealing the capsule as an NFT—without needing to understand the underlying blockchain or storage mechanisms in detail. By presenting each function as a simple user operation rather than as a technical process, the prototype can focus on verifying whether the service concept is understandable, usable, and trustworthy from the user's point of view.

The six operations organized in this document cover the minimum set of actions needed to evaluate the core value of the service. Wallet connection provides a low-friction entry point and also serves as the basis for identity verification. Viewing data information gives users a way to understand the current status of their capsules without exposing sealed content. Registering data and uploading encrypted content form the central user experience, because they are the moments when personal memories, messages, or records are transformed into protected digital assets. NFT generation then finalizes the capsule and provides an immutable proof that the content has been sealed. Finally, access management clarifies how group use, permissions, and exceptional opening conditions should be handled.

A major design principle identified through this research is that the UI should make irreversible actions especially clear. In ChainCapsule, sealing a capsule means that its content can no longer be changed. This gives the service its reliability, but it also creates a responsibility to prevent accidental finalization. For that reason, confirmation screens, clear status labels, registration progress indicators, and warnings about unchangeable settings are essential parts of the minimal UI. These elements are not merely optional usability improvements; they directly support the user's trust in the system.

The discussion on screen priority also shows that the first prototype should concentrate on the shortest complete path that demonstrates the service concept. SCREEN-01, SCREEN-02,

SCREEN-04, SCREEN-06, and SCREEN-08 are therefore treated as required screens for Milestone 1, because they support the essential flow from first connection to NFT issuance. Dashboard, manager operations, wallet information, and group management are also important, but they can initially be implemented in a simplified form as long as they allow the prototype behavior to be verified. More complex functions such as approval management and the opening viewer should be examined in greater detail in Milestone 2, where detailed design decisions can be made with fewer assumptions.

Overall, the findings of this report confirm that the B1 output requirements are satisfied. The required operations have been organized into a clear table, and the data flow for each operation has been described from the user action through to the confirmed result. At the same time, the report identifies practical UI considerations—such as wallet selection, draft editing, upload progress, permission control, and final confirmation—that should be carried forward into the next milestone. These points provide a stable foundation for Milestone 2, where the prototype can move from basic operation definition to detailed screen design and implementation specifications.